

Problem 1

Let $\mathcal{F}$ and $\mathcal{G}$ be σ -fields of subsets of Ω .

- (a) Use elementary set operations to show that $\mathcal{F}$ is closed under countable intersections; that is, if $A_1, A_2, \dots$ are in $\mathcal{F}$, then so is $\bigcap_i A_i$.
- (b) Let $\mathcal{H} = \mathcal{F} \cap \mathcal{G}$ be the collection of subsets of Ω lying in both $\mathcal{F}$ and $\mathcal{G}$. Show that $\mathcal{H}$ is a σ -field.
- (c) Show that $\mathcal{F} \cup \mathcal{G}$, the collection of subsets of Ω lying in either $\mathcal{F}$ or $\mathcal{G}$, is not necessarily a σ -field.

Proof.

- (a) :Let $A_1, A_2, \dots \in \mathcal{F}$. Then $A_1^c, A_2^c, \dots \in \mathcal{F}$.

Since $\mathcal{F}$ is a σ -field, we have $\bigcup_i A_i \in \mathcal{F}$.

By De Morgan's Law, $\left(\bigcap_i A_i\right)^c = \bigcup_i A_i^c \in \mathcal{F}$.

Hence, $\bigcap_i A_i \in \mathcal{F}$.

- (b) :Let $A_1, A_2, \dots \in \mathcal{H}$. Then $A_1, A_2, \dots \in \mathcal{F}$ and $A_1, A_2, \dots \in \mathcal{G}$.

Since $\mathcal{F}$ and $\mathcal{G}$ are σ -fields, we have $\bigcup_i A_i \in \mathcal{F}$ and $\bigcup_i A_i \in \mathcal{G}$.

□

Problem 2

Let $A_1, A_2, \dots$ be a sequence of events. Define

$$B_n = \bigcup_{m=n}^{\infty} A_m, \quad C_n = \bigcap_{m=n}^{\infty} A_m$$

Clearly $C_n \subseteq A_n \subseteq B_n$. The sequences $\{B_n\}$ and $\{C_n\}$ are decreasing and increasing respectively with limits

$$\lim B_n = B = \bigcap_n B_n = \bigcap_n \bigcup_{m \geq n} A_m, \quad \lim C_n = C = \bigcup_n C_n = \bigcup_n \bigcap_{m \geq n} A_m$$

The events B and C are denoted $\limsup_{n \rightarrow \infty} A_n$ and $\liminf_{n \rightarrow \infty} A_n$ respectively. Show that

- (a) $B = \{\omega \in \Omega : \omega \in A_n \text{ for infinitely many values of } n\}$,
- (b) $C = \{\omega \in \Omega : \omega \in A_n \text{ for all but finitely many values of } n\}$,

We say that the sequence $\{A_n\}$ converges to $A = \lim A_n$ if B and C are the same set A .

Suppose that $A_n \rightarrow A$ and show that

- (c) A is an event, i.e. $A \in \mathcal{F}$,
- (d) $\mathbb{P}(A_n) \rightarrow \mathbb{P}(A)$.

Problem 3

Let $\mathcal{E}$ be the left open right closed intervals of $\Omega := \mathbb{R}$ defined in class. Write down the algebra generated by $\mathcal{E}$, and prove your result.

Problem 4

Let Ω be a sample space. Show that any finite algebra on Ω is a σ -algebra.